

4.1

Algebra 1 · Module 4

linear equation

4.1

Algebra 1 · Module 4

standard form

4.2

Algebra 1 · Module 4

rate of change

4.2

Algebra 1 · Module 4

slope

4.3

Algebra 1 · Module 4

slope-intercept form

4.4

Algebra 1 · Module 4

family of graphs

4.4

Algebra 1 · Module 4

parent function

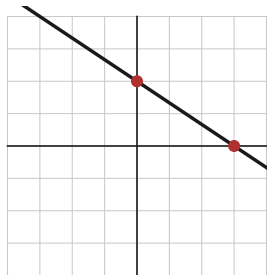
4.4

Algebra 1 · Module 4

identity function

standard form

$Ax + By = C$, where A , B , and C are integers, $A \geq 0$, and A and B are not both 0.



$2x + 3y = 6$: intercepts $(3, 0)$ and $(0, 2)$

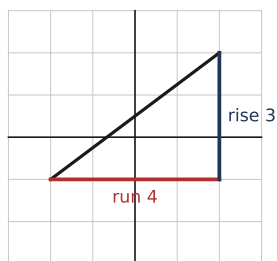
linear equation

An equation whose graph is a line. It can be written in the form $Ax + By = C$.

$$3x - 2y = 6$$

slope

The rate of change of a line: rise over run. $m = (y_2 - y_1) \div (x_2 - x_1)$.



Horizontal: 0. Vertical: undefined.

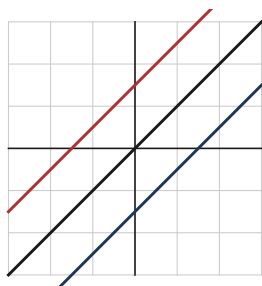
rate of change

How a quantity changes compared with another: change in $y \div$ change in x .

$\$45$ in 3 hours $\rightarrow$ $\$15$ per hour

family of graphs

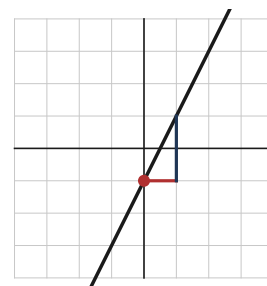
A group of graphs that share one or more common characteristics.



$$y = x, y = x + 1.5, y = x - 1.5$$

slope-intercept form

$y = mx + b$, where m is the slope and b is the y -intercept.



$y = 2x - 1$: start at $(0, -1)$, up 2, right 1

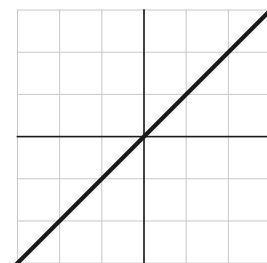
identity function

$f(x) = x$ — every input is paired with itself.

$$f(3) = 3, f(-5) = -5$$

parent function

The simplest function in a family. The linear parent function is $f(x) = x$.



4.4

Algebra 1 · Module 4

translation (graph)

4.4

Algebra 1 · Module 4

dilation (graph)

4.4

Algebra 1 · Module 4

reflection (graph)

4.5

Algebra 1 · Module 4

sequence

4.5

Algebra 1 · Module 4

term

4.5

Algebra 1 · Module 4

arithmetic sequence

4.5

Algebra 1 · Module 4

common difference

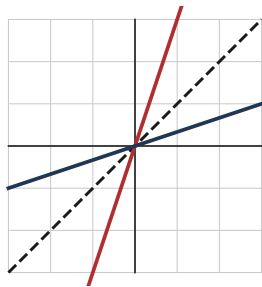
4.5

Algebra 1 · Module 4

**explicit formula (nth
term)**

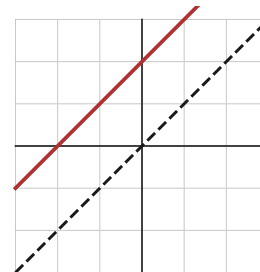
dilation (graph)

Stretches or compresses a graph. $a \cdot f(x)$: steeper if $|a| > 1$, flatter if $0 < |a| < 1$.



translation (graph)

Slides a graph up, down, left, or right without changing its shape.



$f(x) + k$: up k . $f(x - h)$: right h .

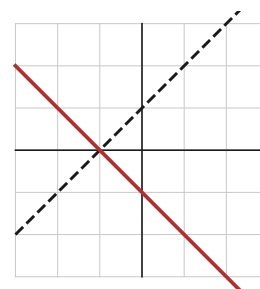
sequence

A list of numbers in a specific order.

3, 7, 11, 15, ...

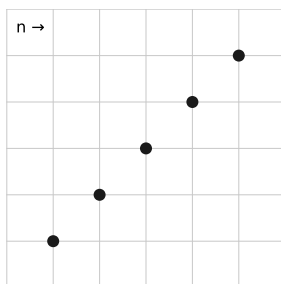
reflection (graph)

Flips a graph over a line.
 $-f(x)$: over the x-axis.
 $f(-x)$: over the y-axis.



arithmetic sequence

A sequence in which the difference between consecutive terms is constant.



Its graph: points on a line (discrete).

term

Each number in a sequence. a_1 is the first term, a_n is the nth term.

In 3, 7, 11, ... the third term is 11.

explicit formula (nth term)

$a_n = a_1 + (n - 1)d$ — finds any term directly.

$$a_1 = 3, d = 4: a_n = 3 + 4(n - 1)$$

common difference

The constant amount added to each term of an arithmetic sequence: $d = a_n - a_{n-1}$.

$$3, 7, 11, 15: d = 4$$

4.5

Algebra 1 · Module 4

recursive formula

4.6

Algebra 1 · Module 4

**piecewise-defined
function**

4.6

Algebra 1 · Module 4

step function

4.6

Algebra 1 · Module 4

**greatest integer
function**

4.7

Algebra 1 · Module 4

**absolute value
function**

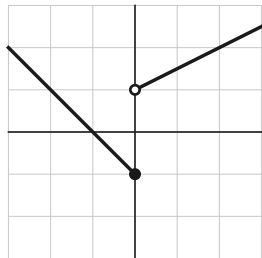
4.7

Algebra 1 · Module 4

vertex

piecewise-defined function

A function with different rules for different parts of its domain.



Open dot: not included.
Closed dot: included.

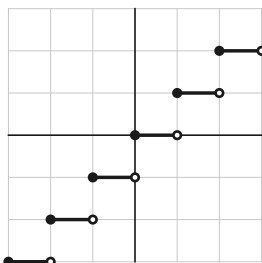
recursive formula

Gives the first term and the rule for getting each term from the one before: $a_n = a_{n-1} + d$.

$$a_1 = 3, a_n = a_{n-1} + 4$$

greatest integer function

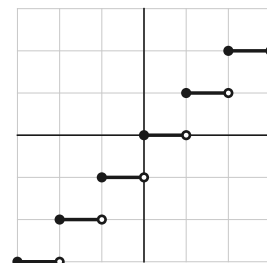
$f(x) = \lfloor x \rfloor$ — the greatest integer less than or equal to x .



$$\lfloor 3.7 \rfloor = 3 \quad \lfloor -1.2 \rfloor = -2$$

step function

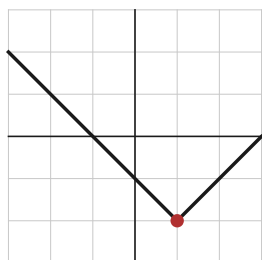
A piecewise function whose graph looks like a set of steps — constant on each piece.



Parking: \$5 per started hour

vertex

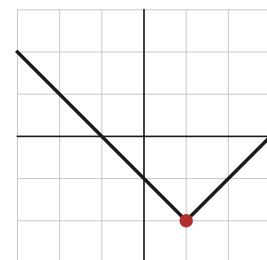
The turning point of the V — the highest or lowest point of an absolute value graph.



$$f(x) = |x - h| + k \text{ has vertex } (h, k)$$

absolute value function

A function with an absolute value in its rule. Parent: $f(x) = |x|$, a V-shaped graph.



$$f(x) = |x - 1| - 2$$